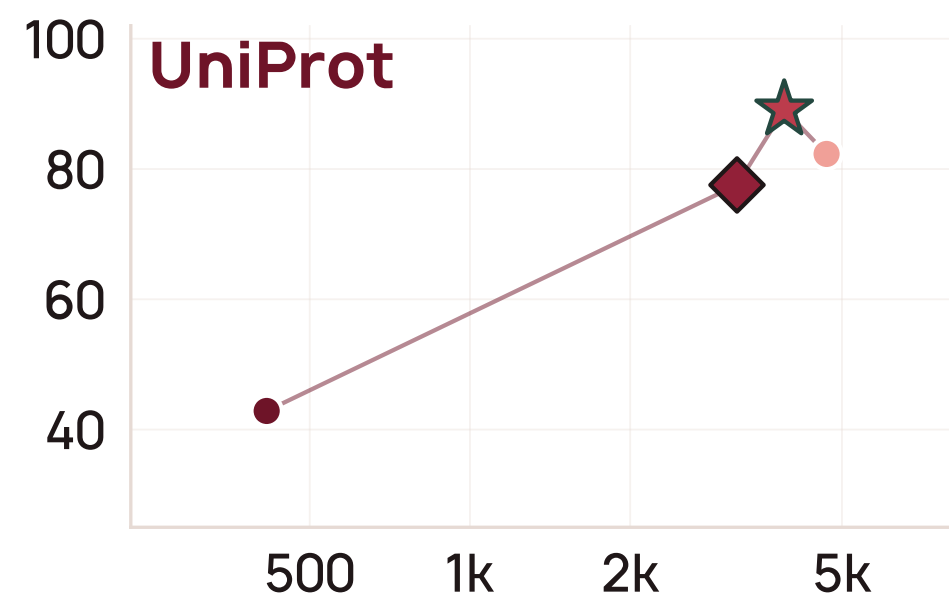
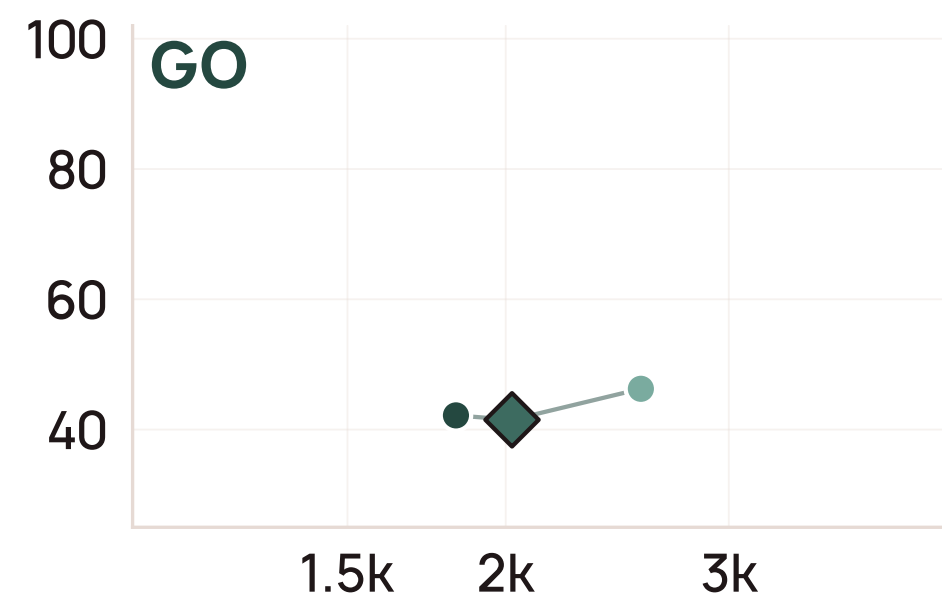


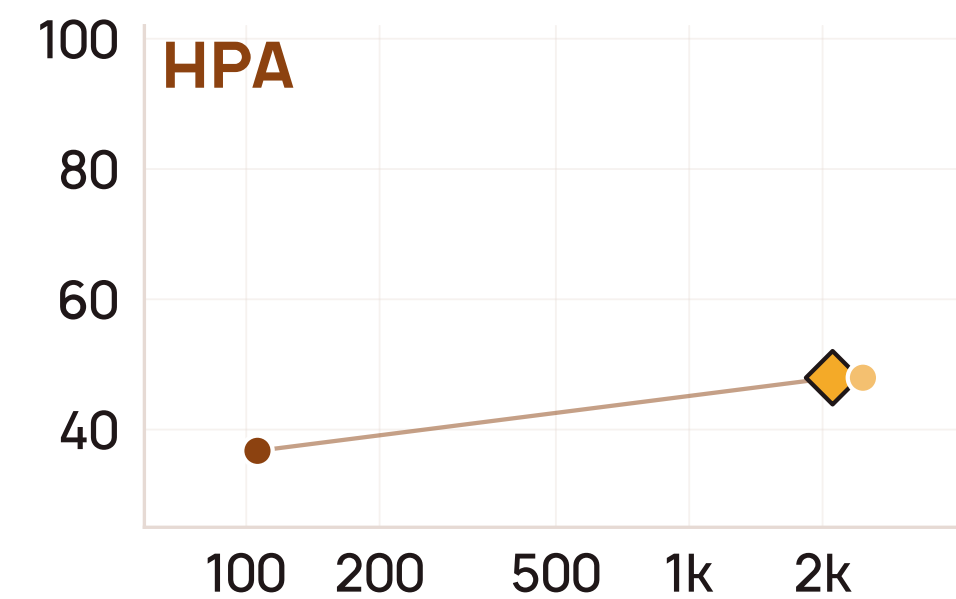
Accuracy on
147-gene benchmark (%)



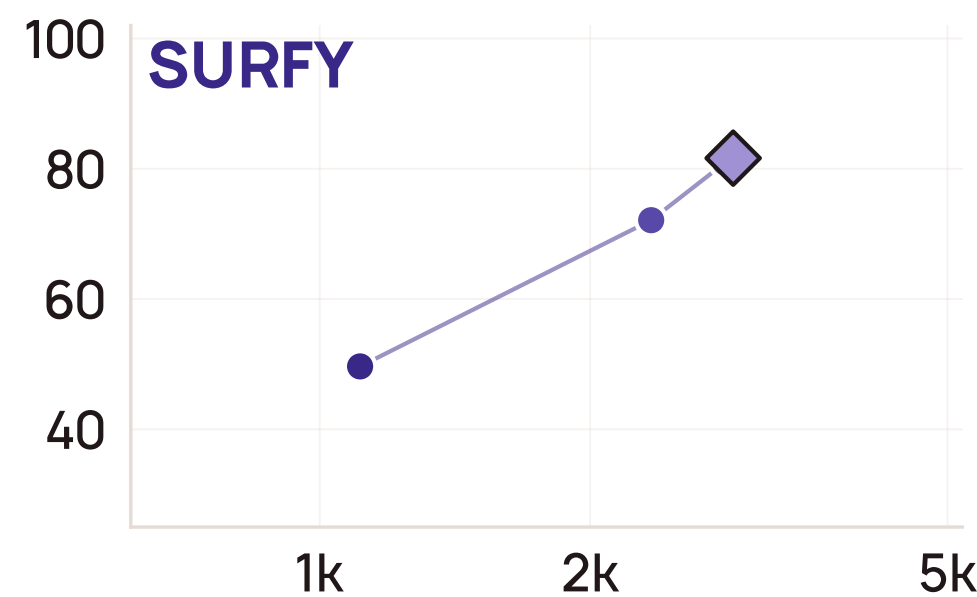
- strict-4 n= 415 +16/-100
- ◆ canonical n=3,175 +69/-96
- ★ TM+signal n=3,894 +87/-94
- permissive n=4,678 +89/-68



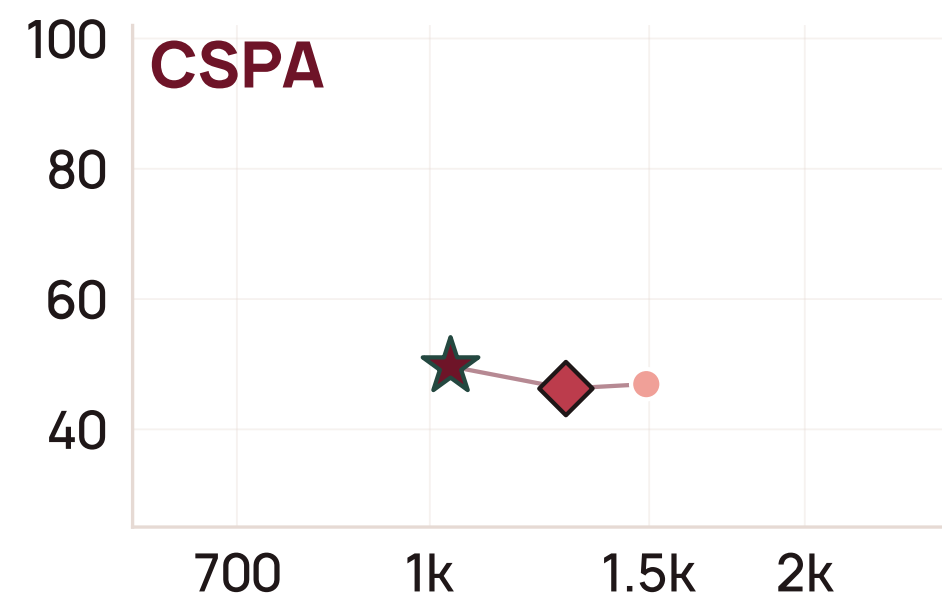
- exp+curated n=1,827 +37/-53
- ◆ canonical n=2,023 +38/-49
- + IEA n=2,557 +47/-45



- Enhanced n= 106 +2/-100
- ◆ canonical n=2,107 +38/-66
- + Uncertain n=2,467 +40/-63



- >0.9 n=1,109 +26/-100
- >0.7 n=2,339 +61/-96
- >0.5 n=2,827 +77/-89
- ◆ canonical n=2,886 +78/-89



- ★ HC-only n=1,039 +37/-77
- ◆ canonical n=1,286 +38/-64
- + unspecific n=1,492 +39/-64

Marker shape

- Alternative cutoff (not used)
- ◆ Canonical (current merge rule)
- ★ Recommended after trade-off audit

Per-source missing rule:
UniProt / GO / SURFY / CSPA absence → predict 'no'.
HPA absence → abstain.

Universe size — proteins this filter would admit (log scale; lower = stricter)